Faeze Mohammadi

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SUMMARY

Computer Engineering student at the University of Tehran with interests across artificial intelligence, machine learning, computer vision, algorithms, and computer systems. Built projects that combine implementation with evaluation and analysis; practiced scientific writing through a Higgs boson review manuscript. Interested in collaborative research grounded in careful reasoning, transparent experiments, and clear communication.

EDUCATION

University of Tehran - Computer Engineering undergraduate **GPA: 19.68 / 20.00**

First-year academic rank: 2nd among Computer Engineering students in the same entering cohort.
Selected course grades (20/20 in each): Probability and Statistics; Discrete Mathematics; Advanced Programming; Artificial Intelligence; Data Structures; Formal Languages and Automata; Engineering Mathematics; Calculus II; Differential Equations.
SAMPAD (NODET) secondary education: Farzanegan 2 and 3, Karaj (2018-2024).
National Olympiad Round 2 qualifier: Math (10-11); Computer Science (10); Economics & Management (10-11).

RESEARCH AND ACADEMIC WRITING

A Study of the Higgs Boson Particle

Sole author | Review manuscript | Available upon request

- Reviewed and synthesized scientific literature on the Higgs boson.
- Organized and wrote the manuscript to present technical ideas clearly in a structured, research-oriented format.

TEACHING EXPERIENCE

Teaching Assistant - Formal Languages and Automata

University of Tehran | Instructor: Dr. Hossein Hojjat

- Supported students with formal models, regular languages, grammars, finite automata, and course problem solving.

Teaching Assistant - Discrete Mathematics

University of Tehran | Instructor: Dr. Siamak Mohammadi

- Helped students work through logic, proofs, relations, combinatorics, graphs, and discrete structures.

SELECTED PROJECTS

Blood Cell Classification with PyTorch GitHub

Compared multilayer perceptron and convolutional network models for blood-cell classification. Examined feature maps, errors, spatial robustness, and occlusion sensitivity to better understand model behavior.

Methods / tools: PyTorch; computer vision; error and robustness analysis.

Supervised Learning from Scratch GitHub

Implemented decision trees and softmax regression, compared them with scikit-learn baselines, and investigated feature engineering and class imbalance.

Methods / tools: Python; supervised learning; model evaluation.

Genetic Music and Minimax Checkers GitHub

Explored music reconstruction with genetic algorithms and adversarial checkers search with Minimax and Alpha-Beta pruning; documented experiments and visual results.

Methods / tools: Optimization; adversarial search; experimental comparison.

PROJECTS CONTINUED

Snake Search with Portals GitHub

Applied BFS, DFS, IDS, A*, and Weighted A* to a portal-based pathfinding problem. Compared how the search methods behave in an interactive setting.

Methods / tools: Graph search; heuristic search; visualization.

Pipelined RISC-V CPU GitHub

Built a five-stage RV32I processor with forwarding, hazard detection, and branch handling. Used program-level tests to verify the pipeline.

Methods / tools: Verilog; computer architecture; verification.

ESP8266 Smart Irrigation GitHub

Created a soil-moisture monitoring system with automatic pump control and a local web dashboard for observing the device.

Methods / tools: ESP8266; embedded systems; local web interface.

Further statistical work: Bayesian robot localization in R with noisy sensing and motion, and Bernoulli Naive Bayes in R for probabilistic spam classification.

COURSES AND CERTIFICATES

Coursera - DeepLearning.AI / Stanford Online

Machine Learning Specialization PDF

- Supervised Machine Learning: Regression and Classification PDF
- Advanced Learning Algorithms PDF
- Unsupervised Learning, Recommenders, Reinforcement Learning PDF

Introduction to Statistics PDF

Writing in the Sciences PDF

OpenCV University

Free Vision Language Models (VLM) Bootcamp PDF

Free TensorFlow-Keras Bootcamp PDF

Free PyTorch Bootcamp PDF

Python for Beginners PDF

Free OpenCV Bootcamp PDF

TECHNICAL TOOLKIT

Programming and hardware: Python, C++, C, R, Verilog, SystemVerilog, HTML/CSS.
AI, ML, and data: PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, NumPy, pandas, Jupyter Notebook.
Systems and tools: RISC-V, FPGA, ModelSim, Quartus, LTspice, Arduino, ESP8266.
Foundations: Algorithms, Data Structures, Probability, Statistics, Formal Languages and Automata, Computer Architecture.

WORKING APPROACH

I prefer projects where the reasoning can be inspected: experiments, verification results, visualizations, and comparisons sit alongside the implementation. I am strengthening my foundations in AI and ML while exploring connections with reliable software and hardware-aware computing.